

Harsh Martinez

Embedded Systems & IoT Firmware Engineer • 7 Years Experience • MERIQ Index: 76% (Strong Candidate)

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TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: C/C++, FreeRTOS, Python, PCB Debugging, MQTT, Embedded Linux, ARM Cortex, I2C/SPI/UART

Verified Certification: Arm Accredited Engineer (AAE)

PROFESSIONAL WORK EXPERIENCE

Senior Embedded Systems & IoT Firmware Engineer — *Bosch* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core C/C++ service, reducing latency by 36%.
- Led a squad of 3 engineers delivering scalable microservices processing 5M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 33%.

Associate Embedded Systems & IoT Firmware Engineer — *Qualcomm* (2019 - 2023)

- Designed and maintained FreeRTOS data endpoints with 97% SLA uptime.
- Optimized database indexing and caching layers using Python saving \$23k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Automotive CAN-Bus Telemetry Gateway with Embedded Linux and Secure Boot [C/C++, FreeRTOS, Python, PCB Debugging]

Engineered end-to-end production solution leveraging C/C++ and FreeRTOS with full test automation.

Smart Grid Power Monitoring Device with MQTT & Hardware Cryptographic Accelerator [Python, PCB Debugging, MQTT, Embedded Linux]

Architected high-resilience system integrating Python with 99.9% uptime.

EDUCATION & CREDENTIALS

M.S. in Robotics and Autonomous Systems

Massachusetts Institute of Technology (MIT) — 2015 - 2019 (Score: 3.71 / 4.0)